

Weekly Engineering Assignment 5 - Group 3 - S23

Manufacturing Data

I don't know because I haven't set a goal yet. If you want to organize, I'll draw a table if the instructions want a table, I'll organize it.

Section 1: Data Processing

Step 1: Reading in the data

Put the rod value according to the first number of the list according to the machine number and the second number of the list according to the month. At this time, load_data declares initialization to zero.

```
In [5]: import csv

# Initialize the rod_data data structure
rod_data = [[[0 for _ in range(20)] for _ in range(3)] for _ in range(5)] # newsted fo

# Read the data from machine.csv and store it in the rod_data data structure
with open('machine.csv', 'r') as file:
    csv_reader = csv.reader(file)
    # Skip the first row (header)
    next(csv_reader)
    n = 0
    for row in csv_reader:
        #rod number will 0 ~ 19 index
        x = n % 20
        n += 1
        machine_num = int(row[0])
        month_num = ['January', 'February', 'March'].index(row[1])
        rod_length = float(row[3])
        rod_data[machine_num][month_num][x] = rod_length

# Print some values from rod_data to confirm that it was read correctly
print(rod_data[1][0][4]) # should print 91.4876798263676
print(rod_data[2][2][19]) # should print 93.2869645905594
print(rod_data[0][0][0]) # should print 91.3779668776973
```

```
91.4876798263676
93.2869645905594
91.3779668776973
```

Step 2: Plot the first machine (Machine 0) data

Graphs corresponding to January can be easily created by using that information is stored in `rod_data[0][0]`. Likewise, other data are just added by calling the right list. Other color line addition lights will be described below.

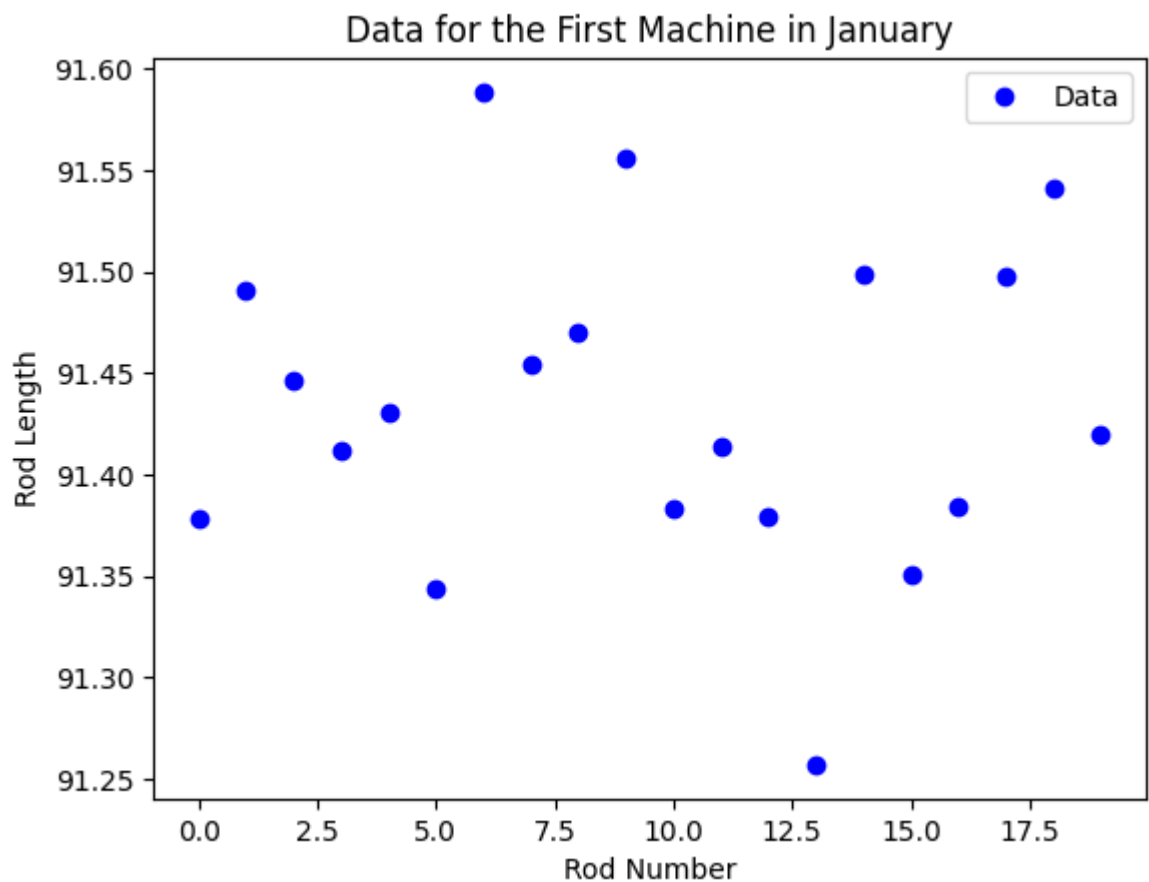
```
In [6]: # Step 1
import matplotlib.pyplot as plt

# Get the data for the first machine in January
jan_data = rod_data[0][0]

# Plot the data
plt.plot(jan_data, 'bo', label="Data")

# Set the title and axis labels
plt.title("Data for the First Machine in January")
plt.xlabel("Rod Number")
plt.ylabel("Rod Length")
plt.legend()

# Show the plot
plt.show()
```



```
In [7]: # Step 2
import matplotlib.pyplot as plt
import numpy as np

# Get the data for the first machine in January
jan_data = rod_data[0][0]
```

```

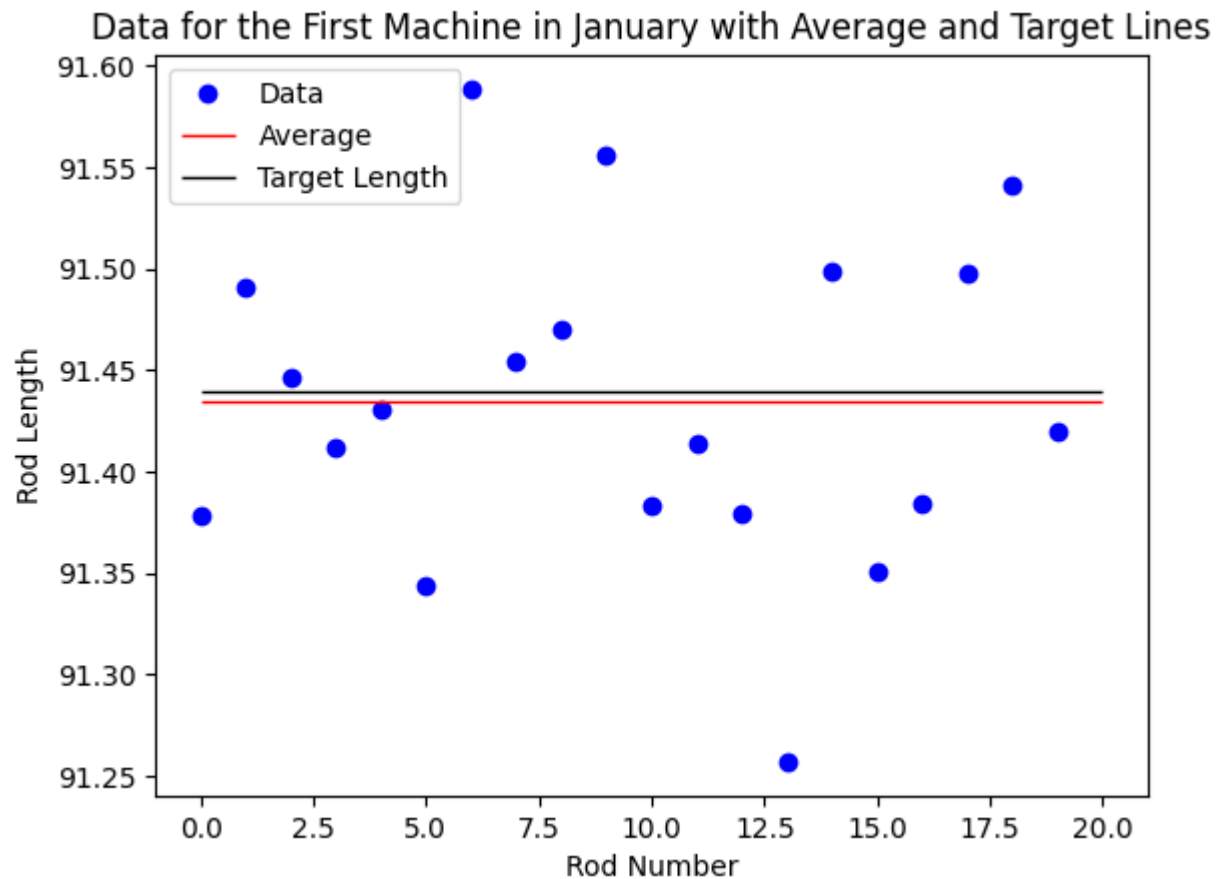
# Compute the average rod length and create a target line
avg_length = np.mean(jan_data)
target_length = 91.44

# Plot the data, average, and target line
plt.plot(jan_data, 'bo', label="Data")
plt.hlines(avg_length, 0, 20, color='red', linestyle='solid', linewidth=1, label = "Av
plt.hlines(target_length, 0, 20, color='black', linestyle='solid', linewidth=1, label

# Set the title and axis labels
plt.title("Data for the First Machine in January with Average and Target Lines")
plt.xlabel("Rod Number")
plt.ylabel("Rod Length")
plt.legend()

# Show the plot
plt.show()

```



```

In [8]: # Step 3: Add all the things
import matplotlib.pyplot as plt
import numpy as np

# Get the data for the first machine for all three months
jan_data = rod_data[0][0]
feb_data = rod_data[0][1]
mar_data = rod_data[0][2]

# Compute the average rod lengths for each month
jan_avg = np.mean(jan_data)

```

```

feb_avg = np.mean(feb_data)
mar_avg = np.mean(mar_data)

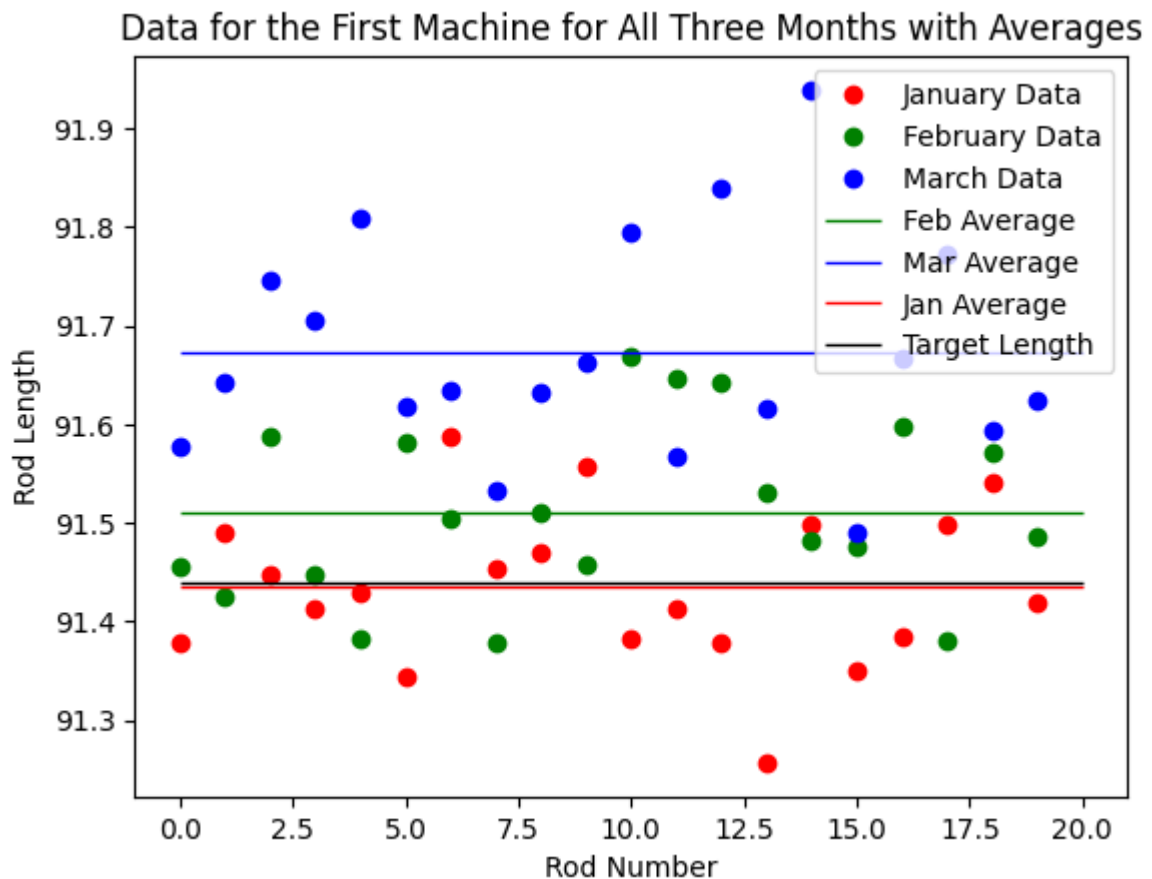
# Create the target lines
target_length = 91.44

# Plot the data and averages for each month
plt.plot(jan_data, 'ro', label="January Data")
plt.plot(feb_data, 'go', label="February Data")
plt.plot(mar_data, 'bo', label="March Data")
plt.hlines(feb_avg, 0, 20, color='green', linestyle='solid', linewidth=1, label = "Feb")
plt.hlines(mar_avg, 0, 20, color='blue', linestyle='solid', linewidth=1, label = "Mar")
plt.hlines(jan_avg, 0, 20, color='red', linestyle='solid', linewidth=1, label = "Jan")
plt.hlines(target_length, 0, 20, color='black', linestyle='solid', linewidth=1, label = "Target")

# Set the title and axis labels
plt.title("Data for the First Machine for All Three Months with Averages")
plt.xlabel("Rod Number")
plt.ylabel("Rod Length")
plt.legend()

# Show the plot
plt.show()

```



It is difficult to analyze the exact value. represent only representative indicators. It is difficult to calculate how much error there is.

Step 3: Plot the third machine (Machine 2) data

```
In [9]: import matplotlib.pyplot as plt
import numpy as np

# Get the data for the thrid machine for all three months
jan_data = rod_data[2][0]
feb_data = rod_data[2][1]
mar_data = rod_data[2][2]

# Compute the average rod lengths for each month
jan_avg = np.mean(jan_data)
feb_avg = np.mean(feb_data)
mar_avg = np.mean(mar_data)

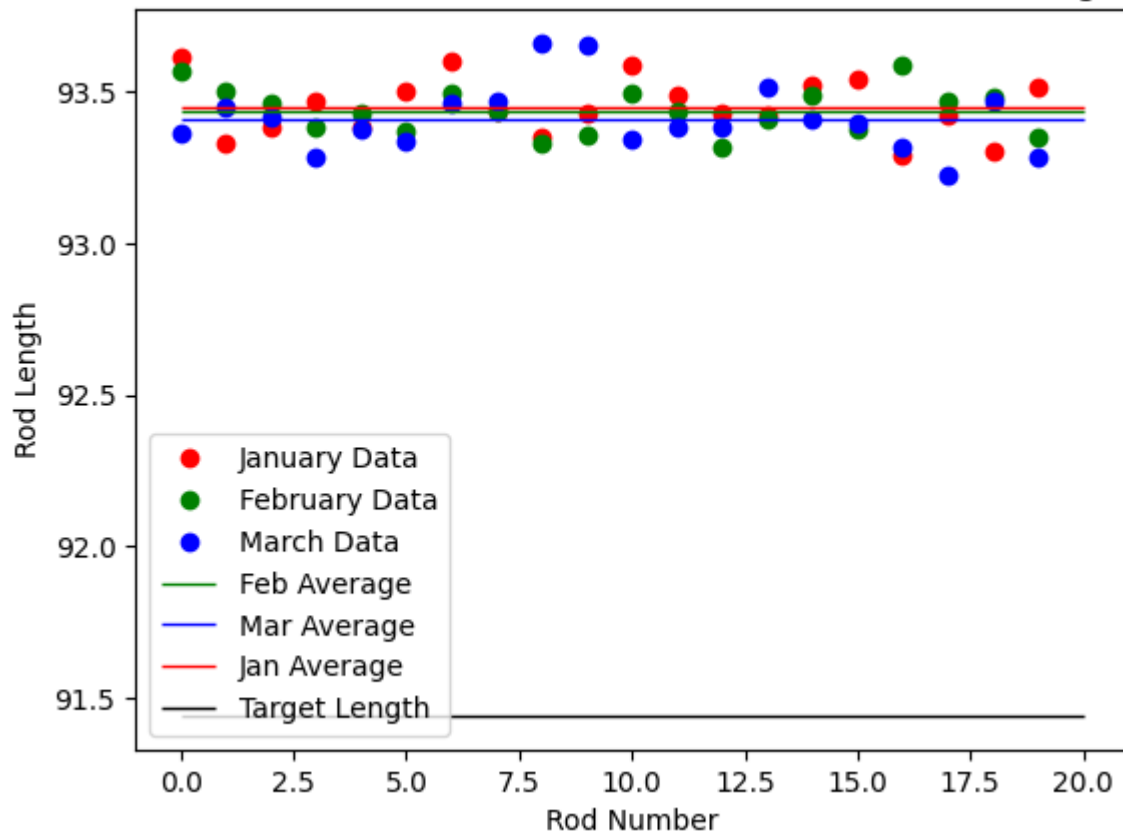
# Create the target Lines
target_length = 91.44

# Plot the data and averages for each month
plt.plot(jan_data, 'ro', label="January Data")
plt.plot(feb_data, 'go', label="February Data")
plt.plot(mar_data, 'bo', label="March Data")
plt.hlines(feb_avg, 0, 20, color='green', linestyle='solid', linewidth=1, label = "Feb")
plt.hlines(mar_avg, 0, 20, color='blue', linestyle='solid', linewidth=1, label = "Mar")
plt.hlines(jan_avg, 0, 20, color='red', linestyle='solid', linewidth=1, label = "Jan")
plt.hlines(target_length, 0, 20, color='black', linestyle='solid', linewidth=1, label = "Target")

# Set the title and axis Labels
plt.title("Data for the 3rd Machine for All Three Months with Averages")
plt.xlabel("Rod Number")
plt.ylabel("Rod Length")
plt.legend()

# Show the plot
plt.show()
```

Data for the 3rd Machine for All Three Months with Averages



On average, it can be seen that it is out of all target values. Mechanical defects may be suspected.

Step 4: Plot the fourth machine (Machine 3) data

```
In [10]: import matplotlib.pyplot as plt
import numpy as np

# Get the data for the thrid machine for all three months
jan_data = rod_data[3][0]
feb_data = rod_data[3][1]
mar_data = rod_data[3][2]

# Compute the average rod lengths for each month
jan_avg = np.mean(jan_data)
feb_avg = np.mean(feb_data)
mar_avg = np.mean(mar_data)

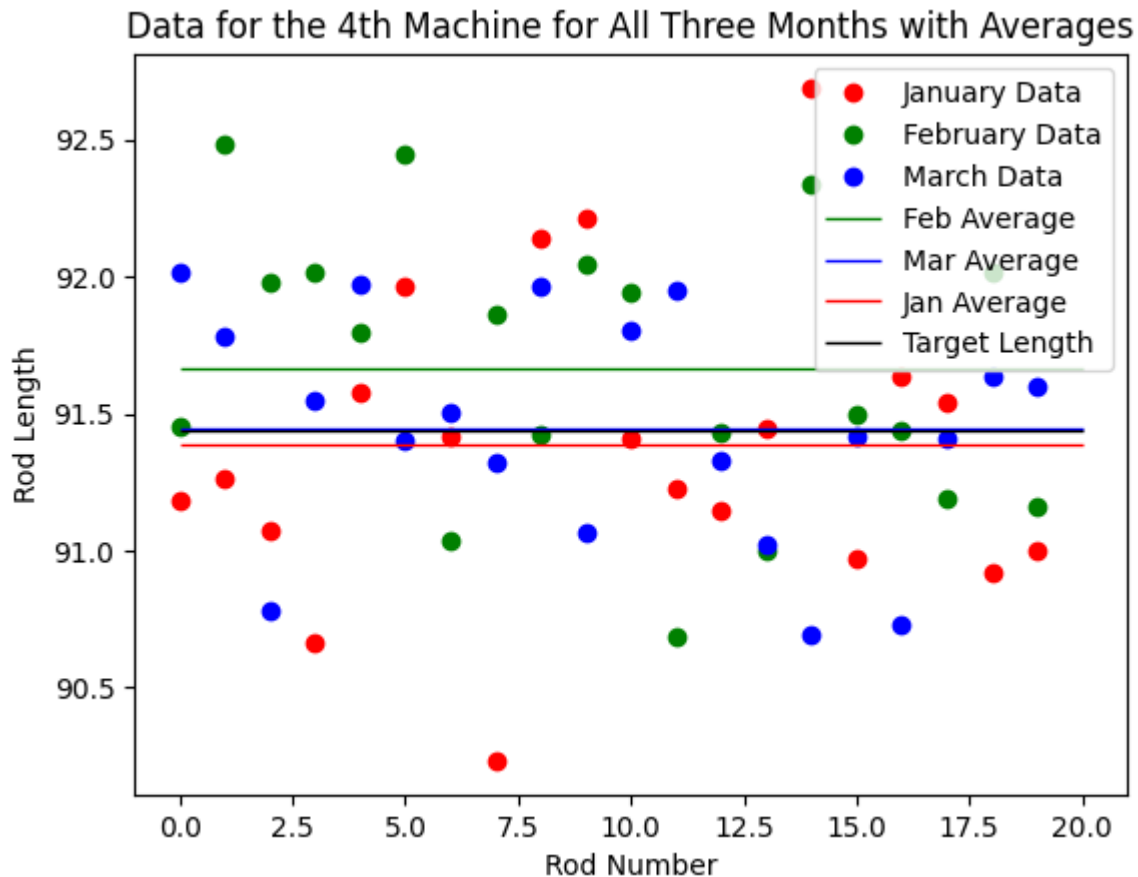
# Create the target Lines
target_length = 91.44

# Plot the data and averages for each month
plt.plot(jan_data, 'ro', label="January Data")
plt.plot(feb_data, 'go', label="February Data")
plt.plot(mar_data, 'bo', label="March Data")
plt.hlines(feb_avg, 0, 20, color='green', linestyle='solid', linewidth=1, label = "Feb")
plt.hlines(mar_avg, 0, 20, color='blue', linestyle='solid', linewidth=1, label = "Mar")
plt.hlines(jan_avg, 0, 20, color='red', linestyle='solid', linewidth=1, label = "Jan A")
```

```
plt.hlines(target_length, 0, 20, color='black', linestyle='solid', linewidth=1, label

# Set the title and axis labels
plt.title("Data for the 4th Machine for All Three Months with Averages")
plt.xlabel("Rod Number")
plt.ylabel("Rod Length")
plt.legend()

# Show the plot
plt.show()
```



There doesn't seem to be much of a problem. February data looks bigger than the average for Target and other months, but the value is actually modest. It would be better to be able to display accurate figures as well as graphs.

Section 2: System Analysis

It is actually easy to express the votes. Since rod_data is well organized at first, it is easy to complete by changing the index in the same form.

```
In [11]: # Jan
import matplotlib.pyplot as plt

# create a List of colors for each machine
colors = ['blue', 'orange', 'green', 'red', 'purple']

# plot the data for all machines in January
```

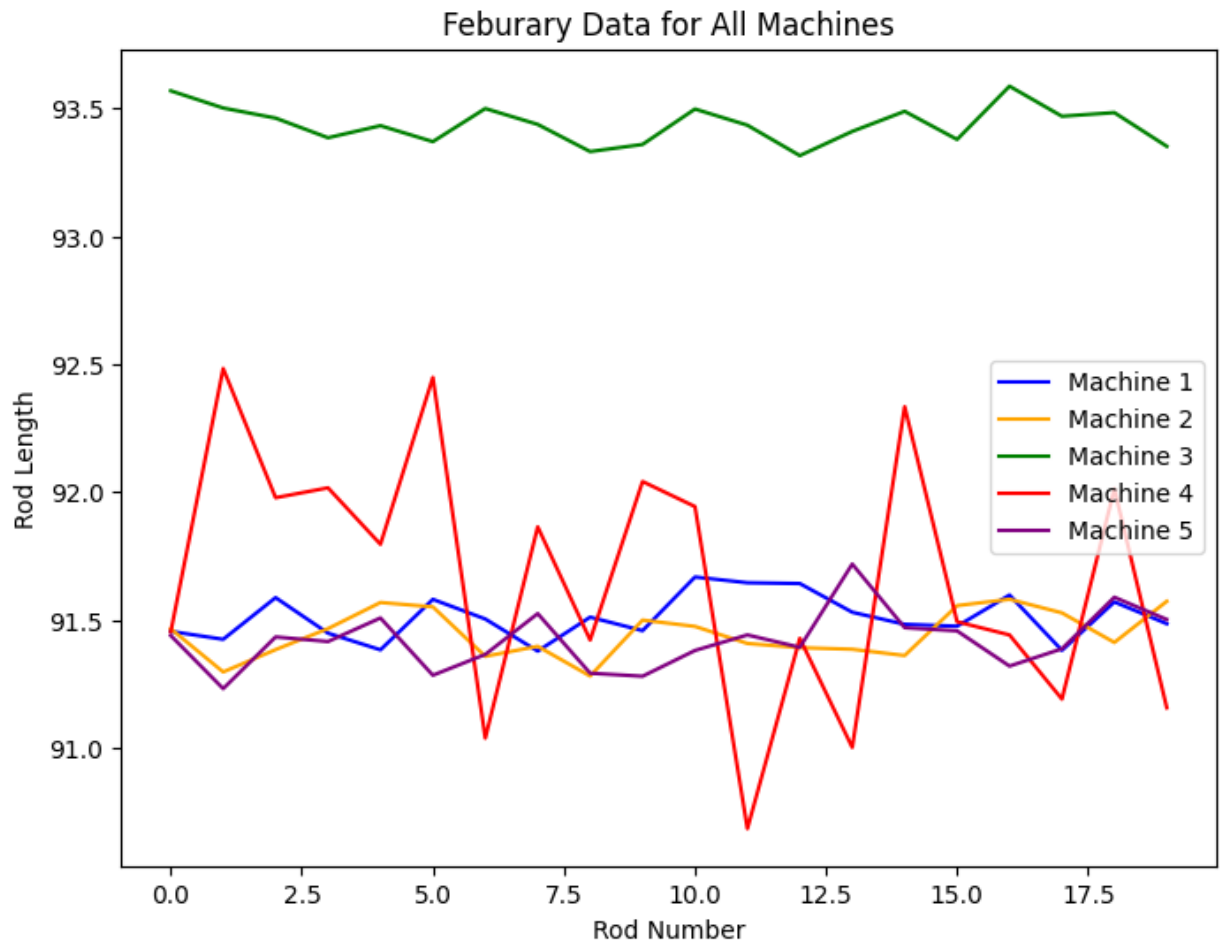
```
plt.figure(figsize=(8,6))
plt.title('January Data for All Machines')
plt.xlabel('Rod Number')
plt.ylabel('Rod Length')
for i in range(1, 6):
    plt.plot(rod_data[i-1][0], color=colors[i-1], label=f"Machine {i}")
plt.legend()
plt.show()
```



```
In [12]: # Feb
import matplotlib.pyplot as plt

# create a List of colors for each machine
colors = ['blue', 'orange', 'green', 'red', 'purple']

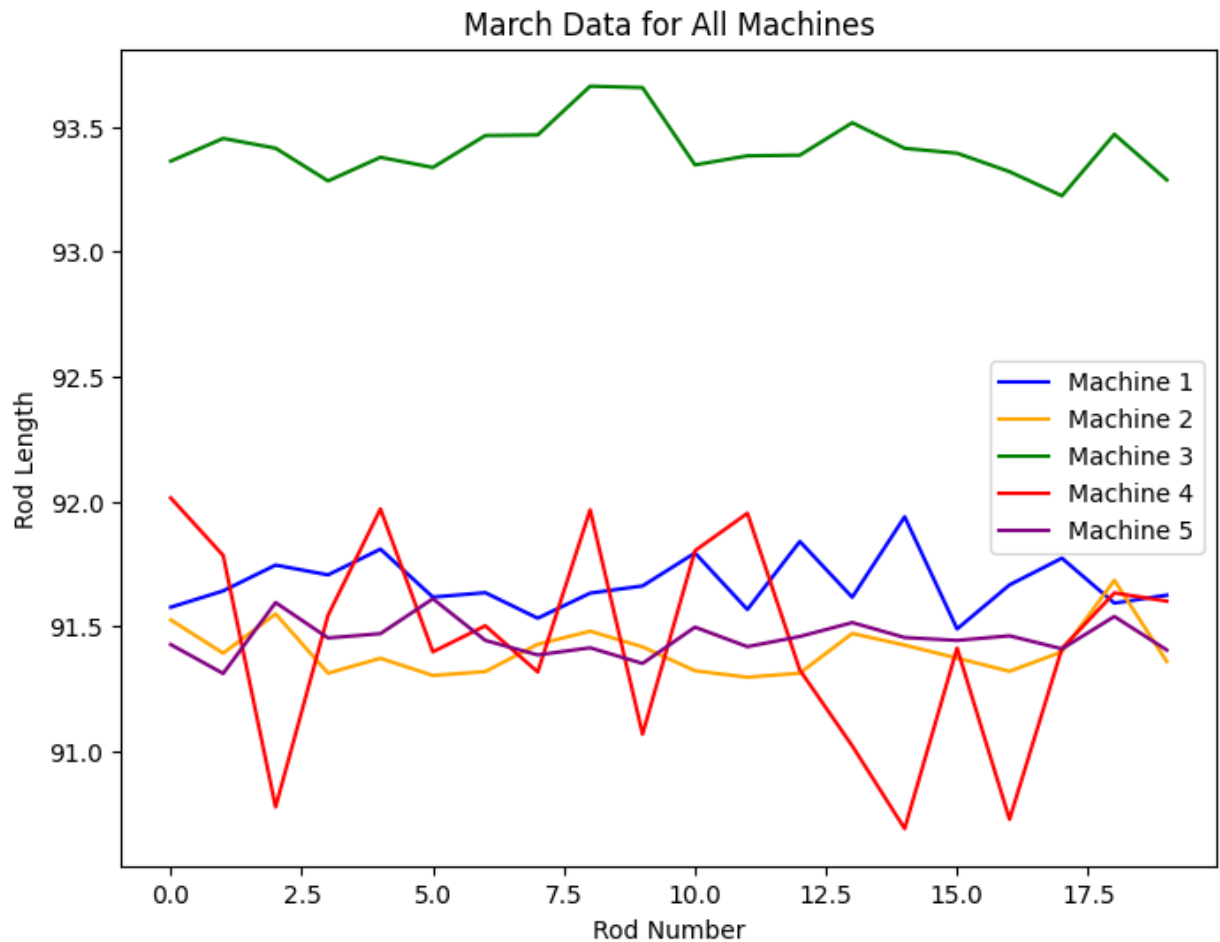
# plot the data for all machines in January
plt.figure(figsize=(8,6))
plt.title('Feburary Data for All Machines')
plt.xlabel('Rod Number')
plt.ylabel('Rod Length')
for i in range(1, 6):
    plt.plot(rod_data[i-1][1], color=colors[i-1], label=f"Machine {i}")
plt.legend()
plt.show()
```

```
In [13]: # Jan
import matplotlib.pyplot as plt

# create a list of colors for each machine
colors = ['blue', 'orange', 'green', 'red', 'purple']

# plot the data for all machines in January
plt.figure(figsize=(8,6))
plt.title('March Data for All Machines')
plt.xlabel('Rod Number')
plt.ylabel('Rod Length')
for i in range(1, 6):
    plt.plot(rod_data[i-1][2], color=colors[i-1], label=f"Machine {i}")
plt.legend()
plt.show()
```



We were provided with information that the target was 91.44, but we did not receive detailed information about the tolerance value. It is certainly a problem that the average value of machine3 is higher than that of other machines. However, this cannot be judged prematurely because the amount of information given is not enough. The same is true of other machines, and it cannot be pointed out unconditionally that the fluctuation is large. If the tolerance range is ± 5 , all machines are normal

Section 3: Data Simulation (Adv.)

Write a function that meets the requirements first. and create the new file. lastly, print out to check it.

```
In [32]: def create_data(num_machines, operators, months, num_measurements, means, stds, filename):
    with open(filename, 'w', newline='') as file: # firstly read the file
        # Create a CSV writer object
        writer = csv.writer(file)
        # Write the header row
        writer.writerow(['machine', 'month', 'operator', 'length'])
        # Loop over the machines
        for machine in range(num_machines):
            operator = operators[machine]
            for month in months: # Loop over the months
                measurements = np.random.normal(means, stds, size=num_measurements) #
                # Write each measurement to the file
```

```
        for measurement in measurements:
            writer.writerow([machine, month, operator, measurement])

# Generate the simulated data
create_data(num_machines=5, operators=['A', 'B', 'C', 'D', 'E'],
            months=['January', 'February', 'March'], num_measurements=12,
            means=91.44, stds=2.45, filename='machine_simulated.csv')

# Read in the file and print out the first and last 5 lines
print("machine data")
with open('machine.csv', 'r') as file:
    csv_reader = csv.reader(file)
    header = next(csv_reader)
    data = list(csv_reader)
    print(header)
    for i in range(5):
        print(data[i])
    for i in range(-1, -6, -1):
        print(data[i])

print("\nMachine_simulated data")
with open('machine_simulated.csv', 'r') as file:
    csv_reader = csv.reader(file)
    header = next(csv_reader)
    data = list(csv_reader)
    print(header)
    for i in range(5):
        print(data[i])
    for i in range(-1, -6, -1):
        print(data[i])
```

```
machine data
['Machine', 'Month', 'Operater', 'Length']
['0', 'January', 'Rosita', '91.3779668776973']
['0', 'January', 'Rosita', '91.49098360970068']
['0', 'January', 'Rosita', '91.44664944554084']
['0', 'January', 'Rosita', '91.4122065529141']
['0', 'January', 'Rosita', '91.43039849450587']
['4', 'March', 'Isabel', '91.40548106727691']
['4', 'March', 'Isabel', '91.53972323535639']
['4', 'March', 'Isabel', '91.41032655405178']
['4', 'March', 'Isabel', '91.46243307879662']
['4', 'March', 'Isabel', '91.44398271329929']
```

```
Machine_simulated data
['machine', 'month', 'operator', 'length']
['0', 'January', 'A', '97.03091317554869']
['0', 'January', 'A', '89.38505463863773']
['0', 'January', 'A', '91.81474922796623']
['0', 'January', 'A', '95.39691089138815']
['0', 'January', 'A', '88.55052983387071']
['4', 'March', 'E', '89.5775828341388']
['4', 'March', 'E', '90.53248032321147']
['4', 'March', 'E', '88.81303001514561']
['4', 'March', 'E', '91.94020296710907']
['4', 'March', 'E', '93.90415456367104']
```

This assignment was just a practice. No separate analysis was required. I paid more attention to how to express the graph beautifully. In the case of Step 3, it seems complicated, but it ends when you establish variables one by one. Others are tasks performed in previous tasks.